Criticmate: Stagewise Human–AI Co-Critique in Single-Screen UI Evaluation

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Abstract

AI tools are increasingly used for UI evaluation, yet most treat evaluation as a single-pass, black-box process that limits both effective model reasoning and human involvement. Grounded in Situation Awareness (SA) theory, we reframe single-screen heuristic evaluation of mobile UIs as stagewise human–AI co-critique, structuring evaluation into three editable stages: Perception (what is on the screen), Comprehension (what elements mean and do), and Projection (what problems and fixes follow). We instantiate this framing in *Criticmate*, an interactive system that exposes intermediate reasoning artifacts for intervention. Across offline benchmarks and a controlled user study, we show that stagewise co-critique yields more expert-like and better balanced critiques than single-pass approaches, while supporting higher trust and engagement without reducing perceived autonomy.

CCS Concepts

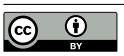
• **Human-centered computing** → **Human-computer interaction (HCI)**; *User interface design*; Empirical studies in HCI; • **Computing methodologies** → *Human-in-the-loop*.

Keywords

Human–AI Collaboration, Heuristic Evaluation, UI Design, Situation Awareness, Human-in-the-loop, Stagewise Co-Critique

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1 Introduction

To rapidly assess whether a user interface is viable, product teams often rely on narrowly scoped methods such as heuristic evaluation and usability inspection. In these approaches, evaluators inspect concrete screens against rule-based guidelines (e.g., Nielsen's heuristics), identifying likely usability problems, visual hierarchy breakdowns, or violations of established conventions [64, 68]. Compared to full-scale user studies, these inspection techniques are lightweight and iterative, allowing teams to catch issues early in the design lifecycle. In this paper, we focus on a common but constrained variant of this practice: heuristic evaluation of individual mobile screens. Designers, students, and stakeholders often exchange static screenshots rather than interactive prototypes and must judge whether a single interface can support a given task based on that static view.

Despite their advantages, heuristic evaluations remain human-intensive and expertise-dependent. The quality and coverage of findings rely heavily on the evaluator's skill; judgments often vary, and repeated inspections across numerous screens impose significant cognitive load and time costs [52]. Practitioners frequently face a trade-off between shallow, checklist-style reviews of many screens and deep inspections of only a few, leaving many design decisions under-examined.

Recent advances in generative models have begun to offload parts of this evaluation work. Metrics can quantify visual properties [71], predictive models can simulate user behaviors [17, 91], and Large Language Models (LLMs) can detect heuristic violations directly from screenshots [17, 85, 93]. Commercial tools such as *Design Buddy* [63] and *Crit Design* [13] now offer immediate AI feedback on mockups.

However, most existing systems model UI evaluation as a *single-pass, black-box* process: the AI produces a static list of comments, and the human can only interpret, filter, or re-prompt after the fact. This formulation imposes limitations on both sides of the collaboration. For the AI, collapsing perception, interpretation, and evaluation into a single step constrains its ability to reason in a grounded, context-sensitive manner, often yielding feedback that is generic,

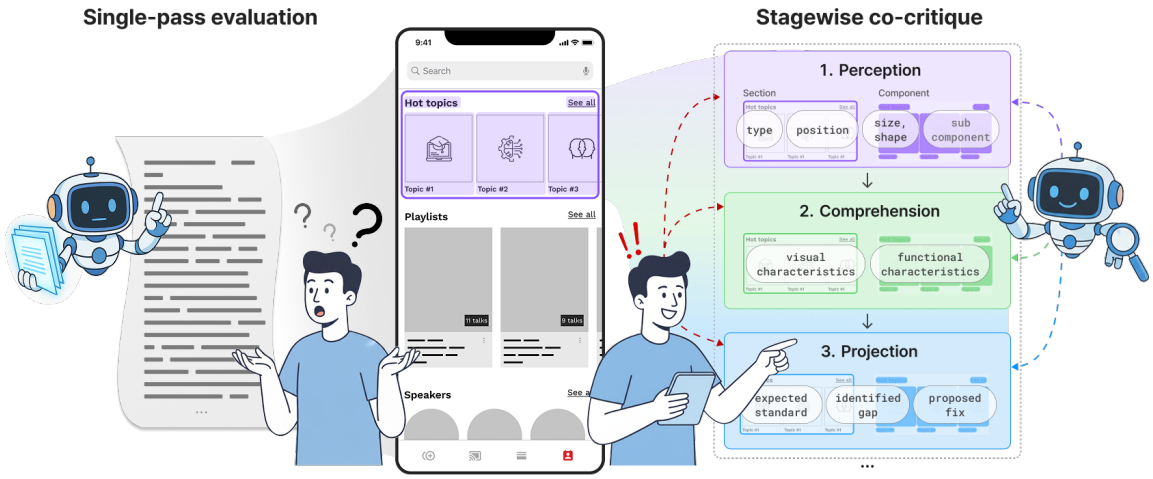


Figure 1: *Criticmate* enables stagewise human–AI co-critique for single-screen UI evaluation, structuring feedback into *Perception* (what is on the screen), *Comprehension* (what it means), and *Projection* (what issues and fixes follow).

weakly anchored in the concrete screen, and difficult to adapt to task or context [17, 48]. For humans, post-hoc interaction with such outputs is opaque and coarse-grained: experts must reverse-engineer the model's reasoning to correct errors, while novices risk automation bias and over-reliance on machine-generated authority [32, 44, 87].

We argue that the core limitation of existing approaches lies not in the availability of automated feedback itself, but in how the evaluation process is structured. When evaluation is treated as a single output, neither the AI nor the human is well supported: the model lacks intermediate scaffolding to reason effectively, and the human lacks clear entry points for understanding and intervention. To address this gap, we reframe single-screen UI evaluation as **stagewise human–AI co-critique**: an approach that decomposes evaluation into a sequence of explicit, editable reasoning stages that jointly support effective model reasoning and meaningful human collaboration.

Drawing on Situation Awareness (SA) theory [21], our formative analysis of UX practitioners' critique practices shows that experts naturally organize evaluation into three cognitive stages: *Perception* (identifying and structuring UI elements), *Comprehension* (interpreting their roles in context), and *Projection* (anticipating usability consequences and formulating remedies). By aligning AI support with this stagewise pipeline, we aim to improve the model's reasoning through structured intermediate representations while simultaneously enabling human collaboration by exposing these stages as explicit points for inspection, correction, and contribution.

We instantiate this paradigm in **Criticmate**, an interactive system that exposes these stages as editable artifacts. In the *Perception* phase, the system maps layout sections and components, which designers can validate and correct. In *Comprehension*, the system analyzes visual and functional characteristics, which designers refine with contextual knowledge. In *Projection*, the system and designer jointly define guidelines, identify gaps, and generate concrete fixes. By giving designers leverage over these intermediate representations, Criticmate scaffolds the AI's reasoning, supports transparent and contestable feedback, and helps mitigate automation bias.

We evaluate Criticmate in two complementary ways. First, using the UICrit dataset (Section 5.2.1) [16] of 1,000 mobile UI screenshots with task descriptions and expert-authored heuristic critiques, we conduct an offline comparison between our stagewise pipeline and single-pass baselines to examine how structuring evaluation into perception, comprehension, and projection affects critique quality. These results show that the stagewise pipeline produces critiques that more closely align with expert feedback and better balance global and local issues.

Second, we conduct a controlled user study with 26 participants (13 professional UX designers and 13 intermediates), comparing Criticmate with a single-pass baseline in realistic critique workflows. Both participants and independent experts rate Criticmate's critiques as more helpful, logically consistent, and concrete, while participants report higher engagement and trust, with comparable perceived autonomy and overall satisfaction.

This work makes four primary contributions:

- **Stagewise Conceptualization of UI Evaluation.** Grounded in Situation Awareness (SA) theory, we conceptualize single-screen UI evaluation as a stagewise process that decomposes evaluation into perception, comprehension, and projection, supporting both effective AI reasoning and explicit points for human understanding and intervention.
- **Criticmate: A Stagewise Co-Critique System.** We design and implement *Criticmate*, an interactive system that operationalizes stagewise human–AI co-critique by exposing intermediate reasoning artifacts as editable representations.
- **Empirical Validation.** We empirically validate stagewise co-critique along two complementary dimensions—AI critique quality and human–AI collaboration—through offline benchmarks and a controlled user study, demonstrating consistent improvements over single-pass approaches.

- **Design Implications.** We distill implications for future human–AI evaluation tools, highlighting how stagewise workflows can support robust model reasoning and collaborative critique across levels of design expertise.

2 Background and Related Work

2.1 Practices of UI Evaluation and Critique

Critique is central to design practice and education, serving as a mechanism for problem detection, intent clarification, and actionable improvements [1, 27]. In studio culture, critique functions as a collaborative space for learning and identity formation [33, 81], making design thinking visible and fostering reflection [89]. Through these practices, designers are socialized into disciplinary reasoning and gradually cultivate expertise [14, 20, 61]. In this broader sense, design critique can engage with conceptual framing, socio-cultural meaning, and long-term implications of design work.

In this paper, however, we focus on a narrower, operational form of critique: single-screen UI evaluation based on heuristic and inspection methods [6]. UI evaluation in this constrained sense is typically conducted through usability inspection methods. *Heuristic evaluation* asks evaluators to inspect interfaces against a set of established principles [67, 68], while *cognitive walkthroughs* and related techniques focus on stepwise task performance and learnability. *Usability testing* captures empirical user behavior and breakdowns in controlled studies [18], and *peer review* or critique sessions foster collaborative learning but depend heavily on evaluators' expertise [35, 79]. Studio formats vary—from *desk critiques* (one-on-one) and *pin-ups* (group discussion) to *final reviews* (collective presentations)—and are often supplemented with techniques such as *think-aloud protocols* or *contextual interviews*, which make designers' reasoning processes more explicit [25, 40]. In industry and education, heuristic evaluation and related inspection methods are widely used as lightweight ways to surface likely usability issues early in the design lifecycle, especially when only static mockups or screenshots are available.

While these approaches provide depth and richness, they are resource-intensive and difficult to scale. In many classrooms and teams, senior designers and external critics have limited time, so only a small number of reviewers can be invited and each project receives only brief attention [37]. Because evaluators bring different tastes and priorities, reviews of the same interface often diverge, a phenomenon known as the "evaluator effect" [36, 39, 45], which makes conflicting feedback difficult to reconcile. Critique situations can also feel socially risky: students and junior designers report feeling exposed when presenting unfinished work and may hesitate to voice disagreement with instructors or senior stakeholders [4, 19]. Differences in expertise further complicate evaluation: experts structure problems at multiple levels and trace issues to broader implications [7, 23], whereas novices focus on surface features or propose unprioritized fixes [97]. Such mismatches affect both feedback quality and receptivity [29], underscoring the importance of systems that not only scale UI evaluation but also adapt to different levels of expertise.

2.2 Automating UI Heuristic Evaluation

Early attempts at automating UI evaluation focused on *measurement* rather than *reasoning*. Metric-based tools such as Aalto Interface Metrics (AIM) quantified visual complexity, exploration time, and color harmony from static mockups [71]. While reproducible at scale, such fixed attributes did not capture design intent, context, or novel patterns. Model-based approaches instead encoded heuristics in design models, rules, or learned predictors to automatically assess usability and heuristic violations [53, 75]. However, these systems were typically engineered and validated for specific domains, and offered explicit support for explaining why a heuristic was violated or how to revise the design. Both metrics and predictive models provided point estimates or labels with little explanatory structure, leaving designers to infer causes and solutions on their own.

More recent work applies large language and multimodal models to generate natural-language feedback for UI heuristic evaluation, including flagging accessibility concerns [2, 85, 95], checking heuristic compliance [17, 59, 93], assessing overall UI quality [65, 90], and simulating user testing with agent personas [8, 58, 92]. For instance, Duan et al. [17] used GPT-4 on Figma mockups and found that while it produced comments on layout clarity and visual hierarchy, its suggestions were often vague and lacked actionable guidance. Similarly, Xiang et al. [92] introduced SimUser, which leveraged LLM-based personas to simulate diverse users (e.g., older adults, novice users) and generate usability feedback overlapping with real user findings.

Despite these advances, current systems fall short of replacing human judgment in UI inspection. Single-pass formulations also constrain AI reasoning by collapsing perception, interpretation, and evaluation into a single step, often leading to brittle grounding and generic feedback. Prior work reports frequent misrecognition of UI structure [34, 94], limited sensitivity to design intent [17, 57], and weakly actionable critiques [48]. From the human side, most systems expose only static outputs, requiring post-hoc re-prompting for refinement. This restricts fine-grained intervention and increases the risk of automation bias. Together, these limitations motivate more process-oriented approaches that better support both model reasoning and human collaboration.

2.3 Human–AI Collaboration in Design

Human–AI collaboration has been widely studied in design, showing how AI can augment creative practice across domains [49]. In graphic design, nonlinear co-design frameworks position AI as an *opinionated colleague* that broadens exploration [96]. In architecture, machine learning systems integrate external aesthetic, environmental, and cultural influences to generate and explore novel architectural proposals [72]. In industrial design, new process models such as the Argyle Design Framework organize iterative divergence–convergence cycles to tame AI-generated idea volume and align generative tools with product design workflows [86], complemented by controlled studies of human–AI co-creation that analyze how users experience collaboration with AI partners [70].

Beyond domain-specific applications, research has characterized broader paradigms of collaboration. Early systems emphasized *passive assistance*, where AI produced static outputs for designers to filter—such as templates or reinforcement-learned product forms

[54]. More recent systems enable *interactive co-creation*, where designers critique AI suggestions and receive refined outputs in response; Shi et al. [84] identify this dialogic cycle across many studies, while Liu et al. [56] show how designers can request nuanced adaptations (e.g., calmer versions for elderly users). Recent work further explores *proactive collaboration*, where AI anticipates designer needs and reshapes collaboration dynamics [12, 56].

Despite these advances, most collaboration efforts remain focused on *output generation*, while evaluation and critique—the meta-level activity of assessing and improving designs—are comparatively underexplored. Fan et al. [26] studied AI-assisted UX evaluation, finding that explanations foster trust and synchronization improves engagement, but their Wizard-of-Oz setup left it unclear how such systems would operate in practice. Current tools remain limited to single-pass evaluations or iterative prompting, producing static heuristic feedback, cumbersome revisions, and risks of automation bias.

We argue that collaboration should extend beyond the *outputs* of UI evaluation toward its *processes*. Instead of classifying systems only by the degree of AI initiative (passive, interactive, proactive), our approach emphasizes how evaluation and critique unfold across stages and clarifies what roles human designers play at each step. By making intermediate reasoning explicit and editable, our proposed method enables designers to engage systematically with perception, comprehension, and projection.

3 Formative Study

3.1 Motivation

To inform the design of a collaborative AI system, we begin by examining how experts actually conduct UI evaluation. Situation Awareness (SA) theory conceptualizes reasoning as a progression through three stages: *Perception* (noticing elements), *Comprehension* (interpreting their meaning), and *Projection* (anticipating consequences) [21, 22]. Although originally developed for safety-critical domains, SA has since been applied to multimodal reasoning and complex decision-making tasks [47, 83]. Prior research shows that when these stages are disrupted, people tend to fixate and explore less effectively; conversely, scaffolding the cycle from perception through comprehension to projection can broaden search and improve outcomes [43, 46, 76].

We adopt SA as an analytical lens to frame UI evaluation as a form of stagewise critique. While prior work has extensively documented *what* usability issues experts identify [35, 52], far less is known about *how* experts reason through these issues across stages of perception, interpretation, and projection [43, 76]. Without a grounded understanding of these processes, systems that impose stagewise structures risk introducing artificial or misaligned boundaries. Accordingly, our goal is to analyze human critique behaviors through the lens of SA and to distill design requirements for stagewise co-critique.

3.2 Methodology

We conducted an in-depth qualitative study with six professional UX practitioners (P1–P6) recruited via an open call in professional UX community chats. Participants were primarily based in industry product teams (five in-house and one freelance), spanning companies across Latin America, Asia, and Africa. They ranged in age from 26 to 37 years ($M = 31.7$, $SD = 4.5$); four identified as men and two as women. Professional UX/UI experience ranged from 1 to 15 years ($M = 5.0$, $SD = 4.8$).

Each participant critiqued two mobile UI screens across two sequential sessions. In the first, they conducted an *unaided critique*, approaching the screen as they normally would in practice. In the second, they performed a *ChatGPT-4o–augmented critique*, examining and revising AI-generated feedback alongside their own reasoning.

To capture both explicit and tacit dimensions of critique, we combined multiple methods. Unaided critiques used a think-aloud protocol to elicit real-time reasoning, while ChatGPT-augmented critiques were accompanied by ethnomethodological observations (e.g., pauses, hesitations, and gestures) to surface implicit decision-making. After both sessions, semi-structured interviews probed how participants perceived interface elements, comprehended their contextual roles, and projected usability consequences.

All sessions were recorded, transcribed, and coded through the lens of SA, with codes organized into perception, comprehension, and projection. Following thematic analysis guidelines [9], three researchers independently coded transcripts and iteratively refined the scheme until reaching consensus.

3.3 Findings

3.3.1 Overall Patterns. Designers' critique practices showed a clear stagewise structure aligned with SA: they identified elements (Perception), interpreted them in context (Comprehension), and projected implications for usability (Projection). By contrast, current AI systems produce single-pass outputs without exposing intermediate reasoning, making it hard for designers to intervene at the right moment.

Designers also fluidly integrated *global* (section-level) and *local* (component-level) perspectives across stages, while AI outputs tended to fragment these views. As critique progressed, participants increasingly relied on domain knowledge, professional heuristics, and contextual awareness, especially when prioritizing issues or proposing remedies. These observations highlight opportunities for AI systems that surface stagewise reasoning, integrate global–local views, and accommodate growing reliance on expert knowledge.

3.3.2 Perception. Designers began by scanning the overall layout to establish the screen's purpose before examining specific components. They relied on global cues such as alignment, spacing, and branding, then shifted to details like button contrast, icon clarity, or text readability. This process was iterative: noticing small irregularities often prompted a return to global interpretation. As P1 explained, *"I start with layout, branding, and general flow. Then I go into detail—contrast, text, buttons."*

AI supported this stage by enumerating visible elements and flagging low-level issues. Some participants found this useful for catching details they might otherwise overlook (P2, P4). However, AI rarely extended its analysis to overall structure or hierarchy, and sometimes omitted central components such as navigation or primary calls-to-action. Participants therefore treated AI output as

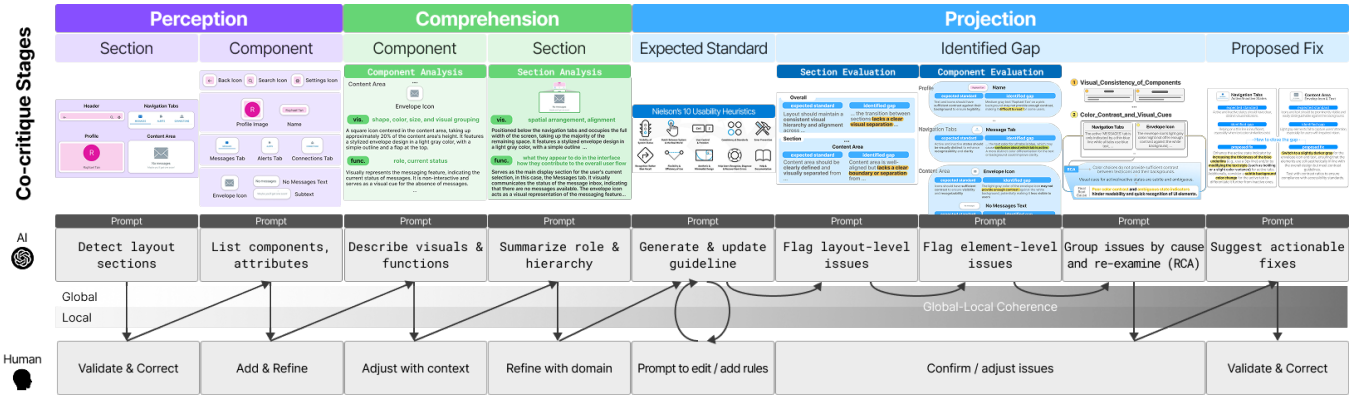


Figure 2: Stagewise co-critique workflow in *Criticmate*. *Perception* builds a structural map, *Comprehension* interprets their visual and functional roles, and *Projection* evaluates issues and proposes fixes using explicit guidelines.

a checklist to be corrected rather than a reliable structural analysis. This gap motivates a Perception stage that explicitly exposes structural recognition for designers to validate or adjust.

3.3.3 Comprehension. After identifying elements, designers interpreted both their *visual characteristics* and their *functional roles* in context. They considered how size, color, alignment, and grouping connected to intended interactions. For example, participants noted that button color suggested priority, or that inconsistent typography weakened information hierarchy. They also reasoned about conventions, label clarity, and task support, often revising interpretations when cues conflicted—for instance, P2 reconsidered a login screen as possibly *ID validation*, while P1 interpreted an icon not as *settings* but *onboarding*. Comprehension further extended to branding, accessibility, and business priorities. As P5 summarized: *"A good structure is when a user can intuitively complete a task without asking for help. The main path should be obvious, the labels should match user expectations, and the system should always give clear feedback."*

AI's comprehension was comparatively weaker. While it sometimes made plausible guesses, its analyses rarely integrated visual and functional aspects. For example, it might describe a button but miss how its color conflicted with brand identity, or label an icon without considering its role in the task flow. Participants described such feedback as generic or redundant, forcing them to supply both contextual and visual reasoning themselves. These shortcomings motivate a Comprehension stage where contextual and visual interpretations can be explicitly integrated and revised.

3.3.4 Projection. In Projection, designers anticipated how current designs would affect user experience in real contexts. They simulated interactions, considered organizational priorities, and evaluated whether the interface would meet user goals under situational constraints. This step required connecting observations to broader values such as accessibility, brand identity, and business requirements.

Their reasoning often resembled *Sadler's formative feedback model* [80]: (1) articulating the *expected standard*, (2) identifying the *gap*, and (3) proposing a remedy. For example, participants judged unclear labels against user expectation principles and proposed revised wording. Beyond this expectation–gap–solution framing, they also engaged in *root-cause reasoning*: tracing clusters of surface-level problems—such as inconsistent spacing, typography, and contrast—back to systemic flaws like the absence of hierarchy rules or style guides. This diagnostic orientation elevated critiques from isolated fixes to strategic, long-term improvements.

By contrast, AI lacked contextual understanding of project goals or organizational constraints, often repeating surface-level issues without linking them to broader usability goals. Participants emphasized that they had to supply such information explicitly and expressed concern about over-reliance if feedback were accepted uncritically. These limitations motivate a Projection stage that connects observations to standards, gaps, and remedies, and supports both surface fixes and root-cause reasoning.

4 Design and Implementation of Criticmate

4.1 System Overview

Criticmate is a web-based client–server system that supports stagewise co-critique. Its design is grounded in our formative study, which showed that designers naturally follow Situation Awareness (SA) stages, integrate global and local perspectives, and inject domain knowledge more heavily in later phases, while current AI tools rarely structure support around these processes.

Figure 2 summarizes the stagewise workflow. Given a UI screenshot and task description as input, Criticmate organizes co-critique into three stages: *Perception*, *Comprehension*, and *Projection*. In *Perception*, the AI segments the layout into sections and enumerates components, and designers validate and adjust this structural map. In *Comprehension*, the AI describes visual and functional characteristics, and designers refine these interpretations using contextual knowledge. In *Projection*, the AI and designers jointly define guidelines, identify gaps, group issues by root cause, and generate fixes. Throughout the workflow, Criticmate maintains explicit links between global and local reasoning and exposes intermediate representations for direct inspection and editing.

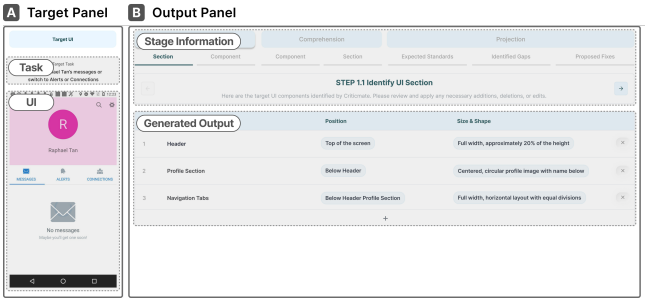


Figure 3: *Criticmate* interface. (A) Target Panel showing the task description and UI screenshot. (B) Output Panel presenting structured, editable AI outputs.

Figure 3 shows the main interface, which consists of two panels. The *Target Panel* (A) displays the task description and the static mobile UI screenshot that serves as the evaluation target. The *Output Panel* (B) shows the current stage of the workflow (e.g., Perception–Section) and the structured outputs generated by the AI, which designers can review and edit. The frontend is implemented in Next.js, and the backend in FastAPI [77] for asynchronous request handling. We use a two-model configuration: GPT-4o (temperature = 0.0) is used for visually grounded analysis in Perception and Comprehension, and o3-mini (temperature = 0.3) is used for Projection, which requires more complex reasoning.

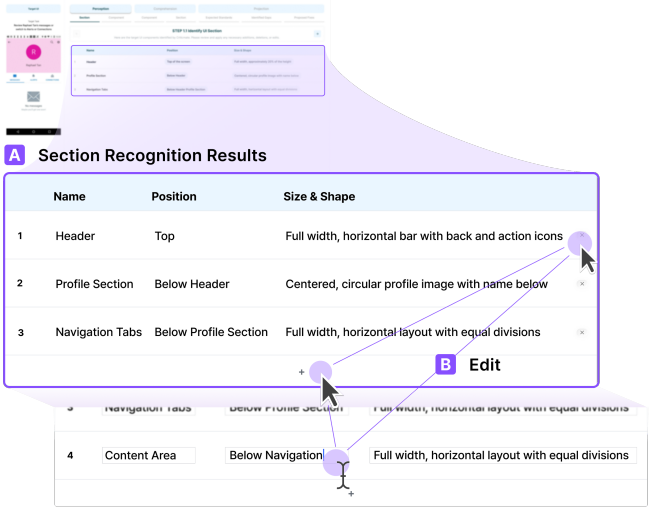
All intermediate results are stored in a structured YAML representation that records sections, components, interpretations, guidelines, gaps, and fixes, enabling serialization and versioned edit history. Concrete YAML examples for each stage are provided in Appendix A, while full-length prompts and implementation details are included in Supplementary Materials, Section S1.

4.2 Perception Stage

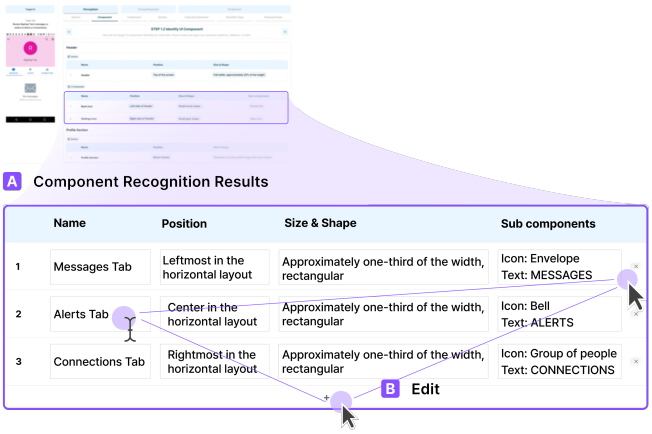
The Perception stage establishes a shared structural foundation by parsing the UI screenshot into a hierarchy of *sections* (layout regions) and *components* (elements within sections). This stage mirrors the global-to-local scanning process observed in our formative study and exposes the model's recognition results as explicit, editable artifacts that designers can validate before higher-level reasoning begins.

4.2.1 Section recognition. As shown in Figure 4a, Criticmate first segments the interface into high-level functional areas (e.g., header, navigation tabs, content grid). The section-recognition prompt enforces exhaustive, non-overlapping coverage and focuses exclusively on the app UI through three constraints: (1) non-app system elements (e.g., the status bar) are explicitly excluded, (2) every remaining pixel must belong to exactly one section, and (3) each section is annotated with spatial attributes (*position* and *size & shape*) in a structured YAML format.

The model's outputs, represented in YAML, are rendered as an editable table in the interface. Designers can rename sections, adjust spatial descriptions, or add or delete rows when the model misclassifies regions or omits relevant layout areas. These edits



(a) Section recognition and editing.



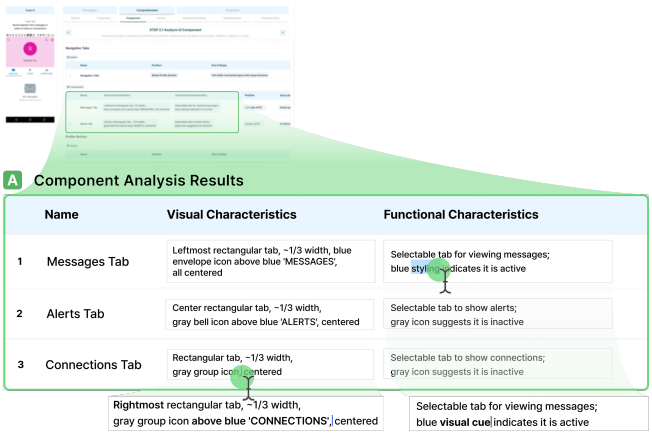
(b) Component recognition and editing within a section.

Figure 4: Perception stage in *Criticmate*. (a) Section-level parsing of the UI into non-overlapping layout regions with editable spatial attributes. (b) Component-level parsing within each confirmed section, exposing individual UI elements as editable artifacts.

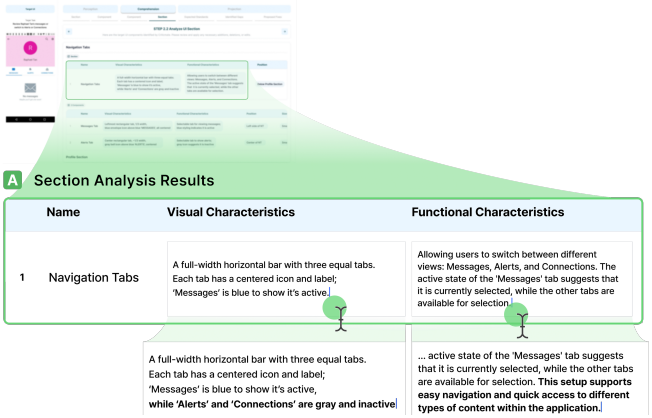
directly update the underlying representation used in subsequent stages.

4.2.2 Component recognition. Given the confirmed section layout, Criticmate next identifies individual UI elements within each section (Figure 4b). The component-recognition prompt instructs the model to recursively enumerate all visible components while explicitly constraining it to avoid hallucinated or inferred elements. The model may not speculate about hidden content or extrapolate patterns from prior knowledge and must remain grounded in visible pixels.

For each component, the model extracts its *component type* (e.g., button, input field), *position*, *size & shape*, and any *sub-components* for grouped elements. These outputs are likewise presented as



(a) Component analysis and refinement.



(b) Section analysis and refinement based on component analyses.

Figure 5: Comprehension stage in *Criticmate*. (a) Component-level interpretations capturing visual and functional characteristics. (b) Section-level summaries aggregating edited component interpretations into higher-level contextual meaning.

editable artifacts. Designers can correct misrecognized elements, refine descriptions, or add missing components before proceeding. The resulting section–component map serves as the shared perceptual grounding for the Comprehension stage.

4.3 Comprehension Stage

Once Perception establishes what is on the screen, the Comprehension stage addresses what those elements *mean* in a single-screen, heuristic-evaluation context. This stage surfaces implicit interpretations and makes them explicit, allowing designers to inspect, refine, and calibrate the model's understanding before evaluative judgments are formed.

4.3.1 Component analysis. As shown in Figure 5a, Criticmate analyzes each component along two explicitly separated dimensions: *visual characteristics* and *functional characteristics*. This separation reflects how designers reason about UI elements and helps prevent premature functional assumptions from being conflated with appearance-based observations.

For the visual dimension, the model describes shape, size, color, layout, spacing, alignment, grouping, and exact on-screen text, as if explaining the component to someone without access to the image. The prompt requires verbatim transcription of visible text (including line breaks and punctuation) and explicitly prohibits inferring semantics from icons or shapes.

For the functional dimension, the model articulates what the component likely does, how it fits into the screen-level task flow, and its apparent state (e.g., selected, disabled), while remaining grounded in visible cues and avoiding unsupported speculation.

In the interface, these interpretations are presented in editable text fields. Designers can refine visual descriptions and adjust functional interpretations to incorporate domain knowledge, branding considerations, accessibility requirements, or business rules.

4.3.2 Section analysis. In addition to component-level interpretations, Criticmate produces section-level analyses that treat groups of components as coherent functional units (Figure 5b). When designers advance to this step, the system uses the edited component descriptions from the previous stage together with the section layout established in Perception to generate an initial summary for each section.

The section-analysis prompt asks the model to describe:

- **Visual characteristics**: overall layout, alignment, flow, visual hierarchy, color and contrast patterns, typography, density, and grouping as observed across components within the section; and
- **Functional characteristics**: what the section appears to do, how it contributes to the stated task and screen-level flow, and its role in the interface structure, based solely on visible evidence and the revised component analyses.

As with component analysis, designers can edit these section-level interpretations to better reflect their intent and contextual understanding. Together, the refined component and section analyses form the semantic grounding that feeds into the Projection stage.

4.4 Projection Stage

The Projection stage translates earlier interpretations into evaluative judgments and actionable recommendations. Whereas Perception and Comprehension focus on recognizing and interpreting what is visible on the screen, Projection addresses what *should* be improved, given a set of expected standards. In our single-screen, heuristic-evaluation setting, this stage makes evaluative reasoning explicit and negotiable by structuring feedback as editable reasoning artifacts.

- **Expected Standard**: the principle or guideline that should be satisfied.
- **Identified Gap**: the observed deviation from that standard.
- **Proposed Fix**: a concrete remedy to close the gap.

This structure reflects how expert designers articulate critiques in practice and helps separate normative expectations from observations and recommendations, reducing ambiguity and overgeneralization.

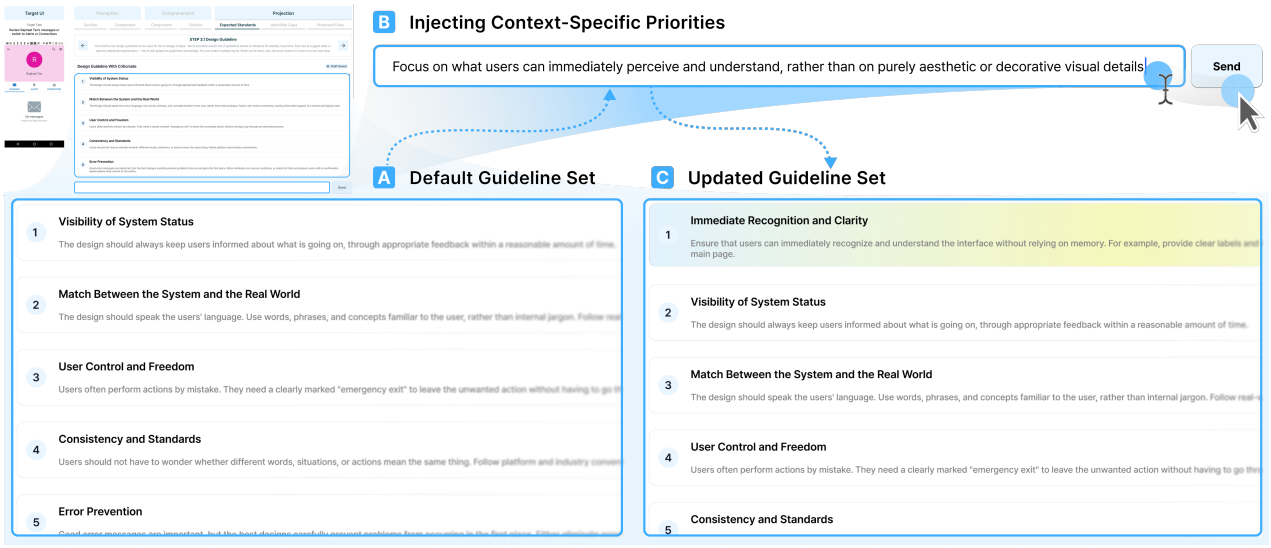
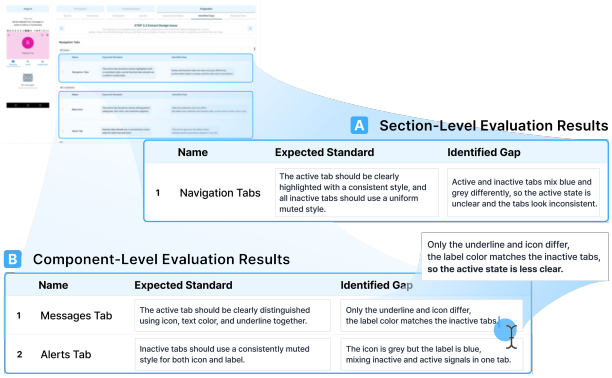
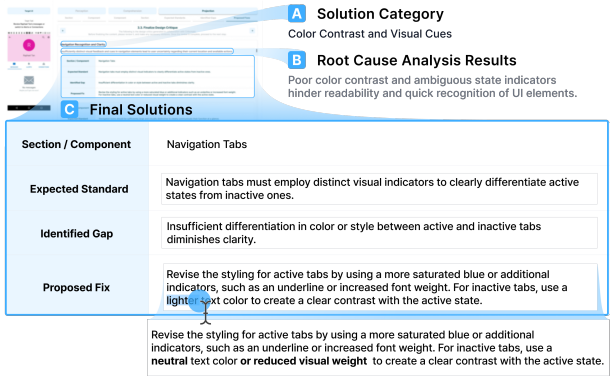


Figure 6: Guideline setup in the Projection stage. Designers extend a default heuristic set with context-specific priorities, which are synthesized into an updated guideline set used as expected standards for evaluation.



(a) Identified gaps at section and component levels.



(b) Root-cause reasoning and proposed fixes.

Figure 7: Projection stage in *Criticmate*. (a) Identification of section- and component-level gaps relative to the guideline set. (b) Grouping of related gaps into root causes and generation of structured solutions as expected standard–gap–fix triplets.

4.4.1 Guideline setup (Expected Standards). The Projection stage begins by establishing the expected standards that will anchor subsequent gaps and fixes (Figure 6). Designers start from a default set of usability heuristics (e.g., Nielsen's heuristics) and then inject task- or domain-specific priorities through free-form notes. Criticmate combines the task description and these notes, passes them through a normalization prompt, and produces a structured, updated guideline list. Designers can iteratively revise this list, using the model as a mediator to synthesize concise and reusable evaluation criteria rather than directly editing guideline text.

4.4.2 Identified gaps: section- and component-level. Given the guideline set, Criticmate next identifies gaps at two levels of granularity (Figure 7a). A first prompt focuses on section-level issues, such as layout structure, ordering, grouping, scanning flow, and visual hierarchy. To maintain stagewise separation, this step explicitly avoids reasoning about labels or individual component functions.

A second prompt targets component-level gaps within each section, focusing on visual clarity, recognizability, consistency, alignment, spacing, and local hierarchy. At both levels, gaps are presented as editable entries linked to the relevant guideline. Designers can rephrase, delete, merge, or add gaps, and can adjust which standards each issue references, directly shaping the evaluative framing.

4.4.3 Root causes and proposed fixes. To move from isolated issues to coherent recommendations, Criticmate then groups gaps into higher-level categories and performs root-cause analysis (Figure 7b). A root-cause prompt receives all section- and component-level gaps

and clusters them based on shared underlying problems, constructing short chains of "why" steps that trace from surface symptoms to deeper causes.

A subsequent solution-generation prompt takes these root-cause categories, the revised guideline set, and earlier Comprehension outputs to propose concrete fixes. This prompt follows a ReAct-style structure (*Thought* → *Action* → *Evaluate Impact*), encouraging stepwise reasoning from diagnosis to intervention while briefly considering side effects for the rest of the screen.

Finally, the resulting Expected Standard, Identified Gap, and Proposed Fix entries are presented as editable fields. Designers can modify, add, or remove any part of the triplet before exporting or sharing the critique.

5 Offline Evaluation of Stagewise AI-Generated Heuristic Critiques

5.1 Objectives

Our overarching goal is to support human–AI collaboration during single-screen heuristic evaluation. Before examining collaboration, however, we analyzed how the newly proposed stagewise structure affects the quality of AI-generated heuristic critiques.

Specifically, we operationalize heuristic evaluation as a three-stage process—Perception, Comprehension, and Projection—and examine how this cognitively motivated structure shapes AI-generated critiques in isolation, independent of human intervention. By doing so, this offline evaluation isolates the effect of stagewise reasoning on the model's behavior itself, providing a foundation for understanding its role in later human–AI collaboration.

We conduct an offline analysis using the UICrit dataset to compare the proposed stagewise pipeline against prompt-based baselines. Our evaluation focuses on properties that matter for expert practice and for downstream use of AI feedback: similarity to expert critiques, balanced coverage of global and local issues, and diversity (non-redundancy) of observations. Specifically, we examine three dimensions:

- **Expert Similarity:** semantic similarity to expert critiques and overlap in referenced UI elements.
- **Global–Local Balance:** whether critiques address both screen-level and component-level issues.
- **Content Diversity:** breadth and non-redundancy of issues and solutions.

5.2 Methodology

5.2.1 Dataset. We used the UICrit dataset [16], which contains 1,000 mobile UI screenshots with task descriptions sampled from Rico [15]. Each UI was annotated by three professional designers, yielding 8,387 expert-authored heuristic critique comments grounded in bounding boxes. To balance interface complexity, we stratified UIs by the number of visible components (`visible-to-user=True`) into four density levels. From each stratum, we sampled 10% to form a held-out test set of 102 UIs. The remaining 898 UIs served as a retrieval pool for baselines that require in-context examples.

5.2.2 Comparison Methods and Ablations. We compare Criticmate against two single-pass baselines and three variants (Figure 8). **ZS** (Zero-shot) generates screen-level heuristic feedback directly from the task description and screenshot in a single-pass [10]. **FS** (Few-shot) additionally retrieves up to eight most similar expert heuristic evaluations from the retrieval pool as in-context examples [16].

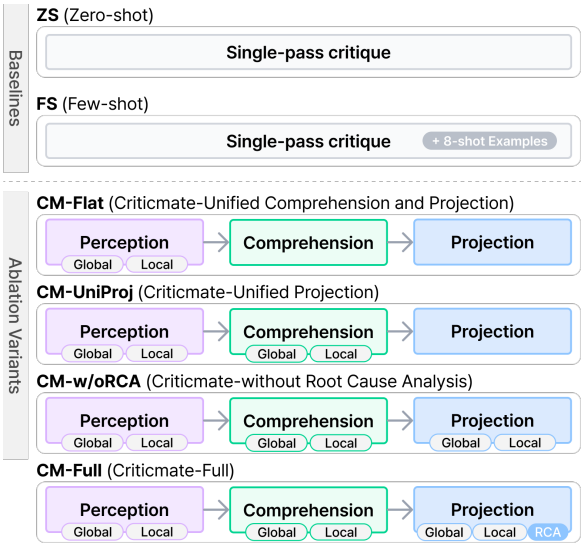


Figure 8: Evaluation conditions for the offline comparison.

To isolate the effect of the stagewise pipeline, we also consider three variants of Criticmate:

- **CM-w/oRCA** (Criticmate without Root Cause Analysis): removes root-cause analysis.
- **CM-UniProj** (Criticmate with Unified Projection): collapses global- and local-level evaluation into a single Projection step.
- **CM-Flat** (Criticmate with Unified Projection and Comprehension): merges Comprehension and Projection, keeping only Perception as a separate stage.

CM-Full denotes the full Criticmate pipeline.

5.2.3 Evaluation Metrics. We assess critique quality using three metric families that capture both expert-likeness and practical utility within our heuristic evaluation setting.

Expert Similarity. We measure similarity to expert heuristic critiques along two dimensions: (1) semantic similarity between comments and (2) whether they refer to overlapping UI elements.

For semantic similarity, we embed all generated and expert comments with Sentence-BERT (`all-mpnet-base-v2` [78]) and compute cosine similarity. A pair is considered a match if similarity ≥ 0.5. We then report:

- **Precision@0.5**: fraction of generated comments that have at least one expert comment with similarity ≥ 0.5.
- **Recall@0.5**: fraction of expert comments that have at least one generated comment with similarity ≥ 0.5.
- **F1@0.5**: harmonic mean of Precision@0.5 and Recall@0.5.

For UI-element overlap, we ask whether semantically matched comments refer to the same parts of the UI. For each expert comment, we select the most similar system comment and extract the set of referenced UI elements from both. An LLM judge (prompt in Supplementary Materials, Section S3.1) determines whether the

element sets overlap (non-empty intersection). We report the proportion of overlapping pairs (1) over all expert comments and (2) restricted to pairs with cosine similarity ≥ 0.5.

Global–Local Balance. Each comment is labeled as Global, Local, or Unclassified by a lightweight LLM classifier (prompt in Supplementary Materials, Section S3.2), applied five times; final labels are by majority vote. To assess internal consistency across runs, we compute Fleiss' Kappa; agreement is very high ($\kappa = 0.935$). We further cross-validate against three human coders. Treating the LLM and three humans as four raters yields excellent inter-rater reliability, quantified as the intraclass correlation coefficient (ICC(2,k) = 0.973; humans only: ICC(2,k) = 0.962), and agreement between the LLM and human majority vote is also high (accuracy = 0.98, Cohen's $\kappa = 0.959$). We then compute, per method, the proportion of comments labeled Global vs. Local.

Content Diversity. We evaluate richness using Comment Count (number of critiques per UI) and Vendi Score [28], a semantic diversity metric computed from SimCSE (Simple Contrastive Learning of Sentence Embeddings) [30] and RoBERTa-base (the base-sized configuration of Robustly Optimized BERT Approach [55]) embeddings. Higher scores indicate broader coverage of distinct issues rather than repetitions. We compute Vendi Scores for all comments and separately for Identified Gaps and Proposed Fixes.

5.3 Results

5.3.1 Expert Heuristic Critique Similarity. Table 1 summarizes semantic similarity to expert critiques at threshold 0.5. Across Recall@0.5, Precision@0.5, and F1@0.5, CM-Full performs best (F1@0.5 = 0.646), with CM-w/oRCA next (0.603) and other methods trailing. Recall@0.5 is consistently higher than Precision@0.5 (e.g., 0.800 vs. 0.542 for CM-Full), reflecting that models generate more comments than experts and thus incur more unmatched outputs. Our diversity results suggest that some "extra" comments may capture additional issues rather than pure noise. Detailed examples of high and low similarity cases are provided in Supplementary Materials, Section S6.

Table 2 complements these findings with UI-element overlap. All methods achieve relatively high overlap (0.817–0.865 over all comments), indicating that high semantic similarity generally corresponds to comments targeting related regions of the UI. However, overlap remains noticeably below 1.0: when a system–expert pair has high semantic similarity but zero overlap, we typically observe (i) a different focus level (e.g., the expert targets a specific button or icon while the model comments on general background or layout), (ii) a different spatial region of the screen, or (iii) different granularity for essentially the same area (e.g., `{"text"}` vs. `{"ad title", "content area", "advertisement"}`). These cases illustrate residual grounding errors that motivate keeping humans in the loop.

5.3.2 Global–Local Balance. Figure 9 shows the proportion of comments classified as Global vs. Local for each method, with mean percentages and 95% bootstrap confidence intervals ($n_{\text{bootstrap}}$ = 10,000; bootstrap details in Supplementary Materials, Section S4). Across overall comments, identified gaps, and proposed fixes, CM-Full consistently produces a substantially larger share of Local comments than any alternative (e.g., ≈39% Local overall, ≈38% for gaps, ≈33% for fixes), while other methods remain in the ≈10–28% range. Non-overlapping confidence intervals between CM-Full and the next-best method (CM-w/oRCA) indicate robust differences rather than sampling noise.

Conversely, baselines and flattened variants skew heavily toward Global comments (often >80%), offering high-level observations but fewer concrete, component-level points. CM-Full therefore achieves a more balanced mix of global and local critique within this single-screen setting, supporting both structural (layout, hierarchy) and element-level (labels, affordances) reasoning in downstream human–AI co-critique. Unclassified comments are rare and do not materially affect these trends.

Table 1: Precision, Recall, and F1 for matching expert critiques, where matches are defined by cosine similarity ≥ 0.5 between AI-generated and expert critique embeddings.

Method	Recall@0.5	Precision@0.5	F1@0.5
CM-Full	**0.800**	**0.542**	**0.646**
CM-w/oRCA	0.708	0.525	0.603
CM-UniProj	0.647	0.383	0.481
CM-Flat	0.615	0.380	0.470
FS (Few-shot)	0.632	0.424	0.508
ZS (Zero-shot)	0.627	0.370	0.465

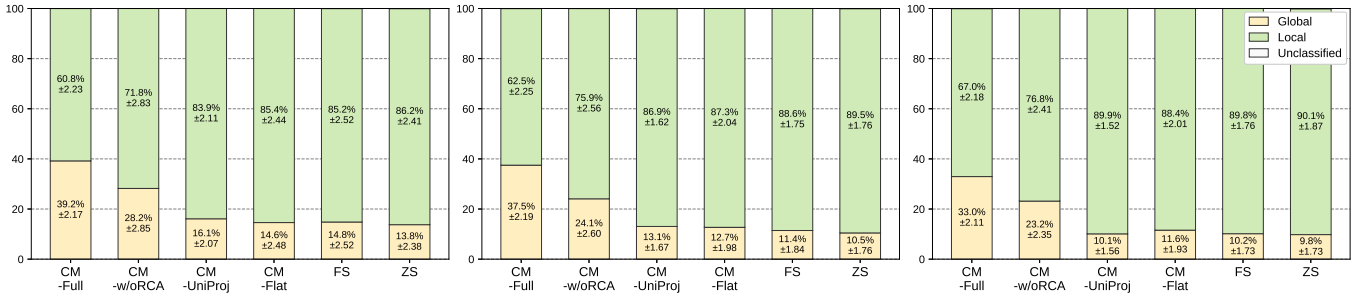
Table 2: UI-element overlap between system-generated critiques and expert annotations. Results are reported over all expert comments and over semantically matched pairs (cosine similarity ≥ 0.5).

Method	all	sim ≥ 0.5
CM-Full	**0.865**	0.896
CM-w/oRCA	0.817	0.868
CM-UniProj	0.838	0.896
CM-Flat	0.860	**0.901**
FS (Few-shot)	0.859	0.896
ZS (Zero-shot)	0.849	0.887

5.3.3 Content Diversity. CM-Full produces both more and more varied comments. On average, it generates 9.6 critiques per UI, compared to 6.1 for FS, 6.5 for ZS, and 5.3–6.4 for the ablations. Without full staged reasoning, important aspects of the interface are often left unaddressed. Figure 10a shows the upward shift in comment counts for CM-Full.

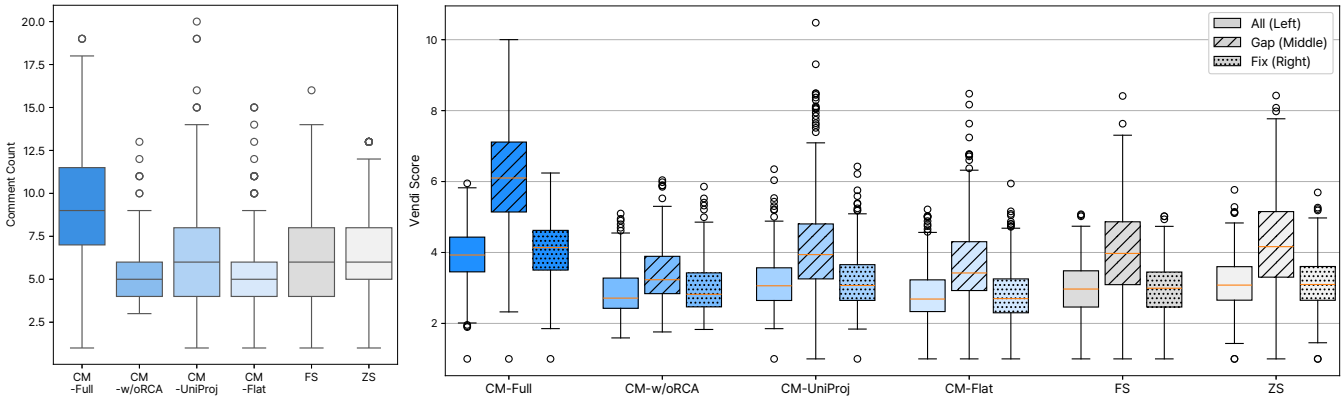
Diversity, measured by Vendi Score, shows an even clearer advantage. CM-Full achieves 3.91 overall, surpassing ZS (3.15) and FS (2.96). For Identified Gaps, it scores 6.07, nearly doubling CM-w/oRCA (3.41) and exceeding FS (4.04) and ZS (4.32). For Proposed Fixes, it reaches 4.06, above all alternatives (best competitor: ZS at 3.16). Figure 10b illustrates this pattern: CM-Full produces more semantically diverse heuristic critiques across categories.

5.3.4 Limitations and Failure Cases . We observed two recurring sources of failure: ambiguous task specifications and underspecified user interfaces. On the task side, generated critiques sometimes



(a) Overall Critique Comments. (b) Identified Gaps. (c) Proposed Fixes.

Figure 9: Global–Local balance of generated critiques across methods. Each bar shows the proportion of comments classified as global vs. local, with error bars indicating 95% bootstrap confidence intervals.



(a) Comment Count per UI & task. (b) Vendi Scores for Overall, Identified Gaps, and Proposed Fixes.

Figure 10: Content diversity of generated critiques across methods. (a) Number of critique comments per UI. (b) Semantic diversity measured by Vendi Score for overall critiques, identified gaps, and proposed fixes.

diverged from expert reasoning when action descriptions were complex or abstract. Tasks combining multiple steps (e.g., *"Enter the mobile number and click on the 'Send OTP' button."* or *"Explore app features like rating, send feedback, and share via available options."*) made it hard to prioritize interactions within a single-screen context. Vague directives such as *"Manage the current sync settings."* similarly left the model uncertain about where to focus. Mismatches also occurred when tasks referenced content not clearly anchored in the UI (e.g., version information or copyright notices). In such cases (e.g., *"View copyrights and version of useful fishing knots."*), Criticmate produced fragmented, tangential, or irrelevant comments, reflecting weak task grounding.

On the UI side, failures arose for screens lacking salient buttons, input fields, or structural landmarks. Cluttered arrangements or ambiguous component placement obscured intended entry points and led Criticmate to misidentify sections or overlook critical elements. We interpret these cases as evidence of the inherent difficulty of reasoning about poorly specified or visually underspecified interfaces in a single-screen setting, and note that such screens are likely challenging even for human experts.

6 User Study with Criticmate

6.1 Objectives

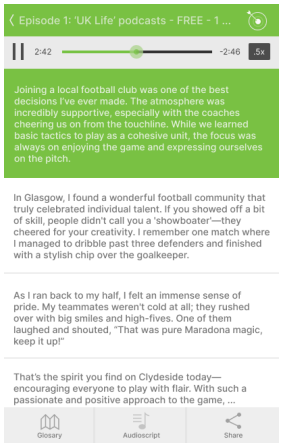
We conducted a user study to compare Criticmate with a single-pass baseline and examine whether its stagewise, SA-guided structure supports *higher-quality single-screen heuristic critiques*, *better collaboration experience*, and *distinct stagewise practices*. We ask:

- **Critique Quality.** How does Criticmate affect the quality of single-screen heuristic feedback?
- **Collaboration Experience.** How does Criticmate shape trust, autonomy, engagement, and satisfaction?
- **Stagewise Practices.** What practices emerge when designers collaborate across stages?

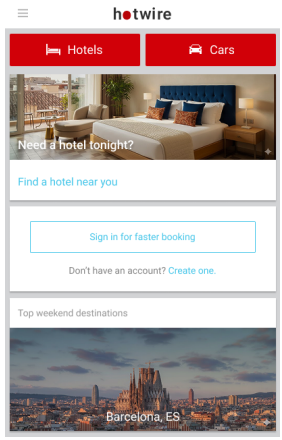
6.2 Participants and Design

6.2.1 Study design and conditions. We used a mixed design. Each participant experienced both collaboration modes (within-subjects):

- **Criticmate**: stagewise, SA-guided human–AI co-critique through Perception, Comprehension, and Projection.
- **Baseline**: single-pass AI critique with iterative natural-language revision.



(a) Information-oriented stimuli (Task: Keep listening to the podcast or pause.).



(b) Task-oriented stimuli (Task: Sign in or create an account to book a nearby hotel.).

Figure 11: UI stimuli and tasks used in the study. Screens were redrawn for publication while preserving key layout and interaction elements.

This baseline reflects common current practice in utilizing automated UI evaluation: designers request a single full critique, then iteratively steer the system with targeted prompts (e.g., "focus on accessibility," "de-emphasize visual nitpicks"), including instructions to revise or delete parts of the AI's prior feedback. Comparing Criticmate against this pattern allows us to ask whether a stagewise, SA-guided process offers benefits beyond this single-pass critique plus ad-hoc steering workflow.

Expertise (Experts vs. Intermediates) served as a between-subjects factor. Order of collaboration modes was counterbalanced within each expertise group. Each participant evaluated two UIs, and the mapping between UIs and modes was counterbalanced so that every system–UI pairing was observed across participants.

6.2.2 Participants. We recruited 26 participants: 13 professional designers with industry UI/UX experience (**Experts**, E; mean industry experience = 3.15 years) and 13 intermediates with project-based but limited professional experience (**Intermediates**, I). Recruitment was through university design departments and professional social media. The study involved no invasive procedures or identifiable data and was classified as minimal risk under institutional guidelines. All participants gave informed consent and received compensation equivalent to USD 30.

Sample size planning followed an *a priori* power analysis with G*Power. Because standard tools do not directly support linear mixed-effects models, we approximated our design as a mixed ANOVA with one within-subject factor (mode: 2) and one between-subject factor (expertise: 2). Assuming a medium effect size (f = 0.25), α = 0.05, and power = 0.8, the analysis suggested at least 24 participants. Our sample of 26 exceeds this bound and aligns with prior HCI work on AI-supported evaluation and critique [11, 17, 88].

6.2.3 Stimuli. We selected UI stimuli from the subset of UICrit (Section 5.2.1) consisting of the top 50% of screens with the largest number of visual elements, in order to ensure sufficient structural and functional complexity for single-screen heuristic evaluation. First, we restricted candidates to screens with many visible components. Second, we aimed to cover structurally distinct interaction contexts that participants commonly encounter in practice. Accordingly, we chose one *information-consumption–oriented* screen and one *task–oriented* screen from this high-density subset (Figure 11). The first stimulus is an information consumption–oriented UI for listening to a podcast, paired with the task description *"Keep listening to the podcast or pause."*. It combines long-form text and audio controls, representing a pattern in which users primarily consume information while intermittently invoking control functions. The second stimulus is a task-oriented login screen, paired with the task *"Sign in or create an account to book a hotel nearby."*. Unlike a purely informational display, this screen requires users to actively enter information and understand a short procedural flow (sign-in vs. sign-up) in order to complete a clearly defined goal.

6.2.4 Baseline condition. In the Baseline condition, participants uploaded the UI and task description. The system invoked the same vision–language model used in Criticmate (gpt-4o) and applied Nielsen's ten heuristics as default guidelines. A single-pass prompt produced a YAML-formatted critique structured as a Sadler-style triplet: `expected_standard`, `identified_gap`, and `proposed_fix`, plus a `component` field linking to the relevant UI region. The full prompt for the baseline condition is provided in Supplementary Materials, Section S2.

Participants could then refine this critique indirectly by entering free-text revision requests. These requests could adjust guidelines (e.g., "emphasize error prevention and recovery") or the output (e.g., "downplay minor visual inconsistencies and focus on task-blocking issues"). Each request, plus the original YAML, was passed to a second prompt that updated the YAML while preserving its structure. Participants could iterate this generate–revise loop freely within a 30-minute session.

Thus, both conditions allowed participants to steer AI-generated critiques, but with different edit loci: the baseline supported global revisions of the final critique via natural-language instructions, while Criticmate exposed intermediate Perception and Comprehension artifacts as directly editable structures before Projection.

6.3 Procedure

Sessions were conducted remotely via video conferencing and screen sharing, and lasted about 90 minutes. After consent and a brief orientation, participants used both systems in counterbalanced order. Each system was introduced via a short tutorial, then used for a 30-minute evaluation session on one UI. In each session, participants used the assigned system to produce single-screen heuristic critiques for the given UI and task, refining or supplementing AI output until they felt it was sufficiently complete. All interactions (system outputs, edits, revision notes) were logged. After both sessions, participants completed post-task questionnaires (Section 6.6.4) and answered open-ended questions about strengths, weaknesses, and reasons for intervening in AI feedback.

6.4 Measures

We evaluated the study along two primary dimensions—*critique quality* and *collaboration experience*–supplemented by qualitative feedback and interaction logs. Here, "critique" refers specifically to participants' written heuristic-style feedback on a *single* screen.

6.4.1 Critique Quality. We measured critique quality via (a) participants' self-assessments and (b) independent expert assessments, using the same four dimensions drawn from prior work on design feedback and critique [17, 31, 60]:

- **Helpfulness**: *"This critique provides practical help for improving the given UI design."*
- **Consistency**: *"The critique's claims are logically consistent with clear reasoning."*
- **Clarity**: *"The critique is easy to understand and clearly expressed."*
- **Concreteness**: *"The critique is specific to elements of the given UI, rather than general advice."*

All items used 7-point Likert scales. For external assessment, three professional designers (**J**) rated each critique independently and blindly: artifacts were anonymized and randomized, and system labels were removed.

6.4.2 Collaboration Experience. We measured participants' experience with each system along four dimensions. *User Engagement* was measured using the 12-item User Engagement Scale–Short Form (UES-SF) [73] on a 5-point Likert scale, with subscales for Focused Attention, Perceived Usability, Aesthetic Appeal, and Reward. *Trust* was measured with three 7-point items adapted from [38]. *Perceived Autonomy* used three 7-point items about the extent to which participants felt their opinions and preferences were reflected [38]. *Satisfaction* was measured with a single 7-point item (*"I am generally satisfied with the generative AI service experience just presented"*) [38]. Full item wordings are in Appendix B.

6.4.3 Qualitative Feedback and Logs. Open-ended survey questions collected participants' reflections on strengths and weaknesses of each mode and their reasons for editing AI feedback. Interaction logs (outputs, edits, revisions) provided behavioral evidence of when and how participants intervened during evaluation.

6.5 Data Analysis

We analyzed quantitative data using linear mixed-effects models (LMMs). In our study design, each participant produced critiques under two conditions (Criticmate vs. Baseline), belonged to one of two expertise levels, and each UI stimulus was reused across conditions. LMMs model participant- and UI-specific variability via random effects, effectively controlling for these sources of variation while estimating the fixed effects of expertise and condition.

For external expert ratings of critique quality, we used:

$$\begin{aligned} \textit{rating} \sim\ & \textit{expertise} * \textit{condition} \\ & + (1 \mid \textit{participant}) + (1 \mid \textit{UI}) + (1 \mid \textit{evaluator}), \end{aligned}$$

where *rating* is an expert's score on one dimension, *expertise* encodes whether the critique author was an Expert or Intermediate, and *condition* encodes Criticmate vs. Baseline. Random intercepts capture variability due to author, stimulus, and external rater.

For participants' self-ratings (perceived critique quality and collaboration experience), we used:

$$\textit{rating} \sim \textit{expertise} * \textit{condition} + (1 \mid \textit{participant}) + (1 \mid \textit{UI}).$$

All models were estimated via restricted maximum likelihood (REML). We obtained p-values and denominator degrees of freedom using Satterthwaite's approximation and applied Holm–Bonferroni correction within each family of related outcomes. Residual diagnostics did not indicate substantial assumption violations. Qualitative data (open-ended responses and logs) were analyzed with thematic analysis and iterative codebook refinement by three researchers.

6.6 Results

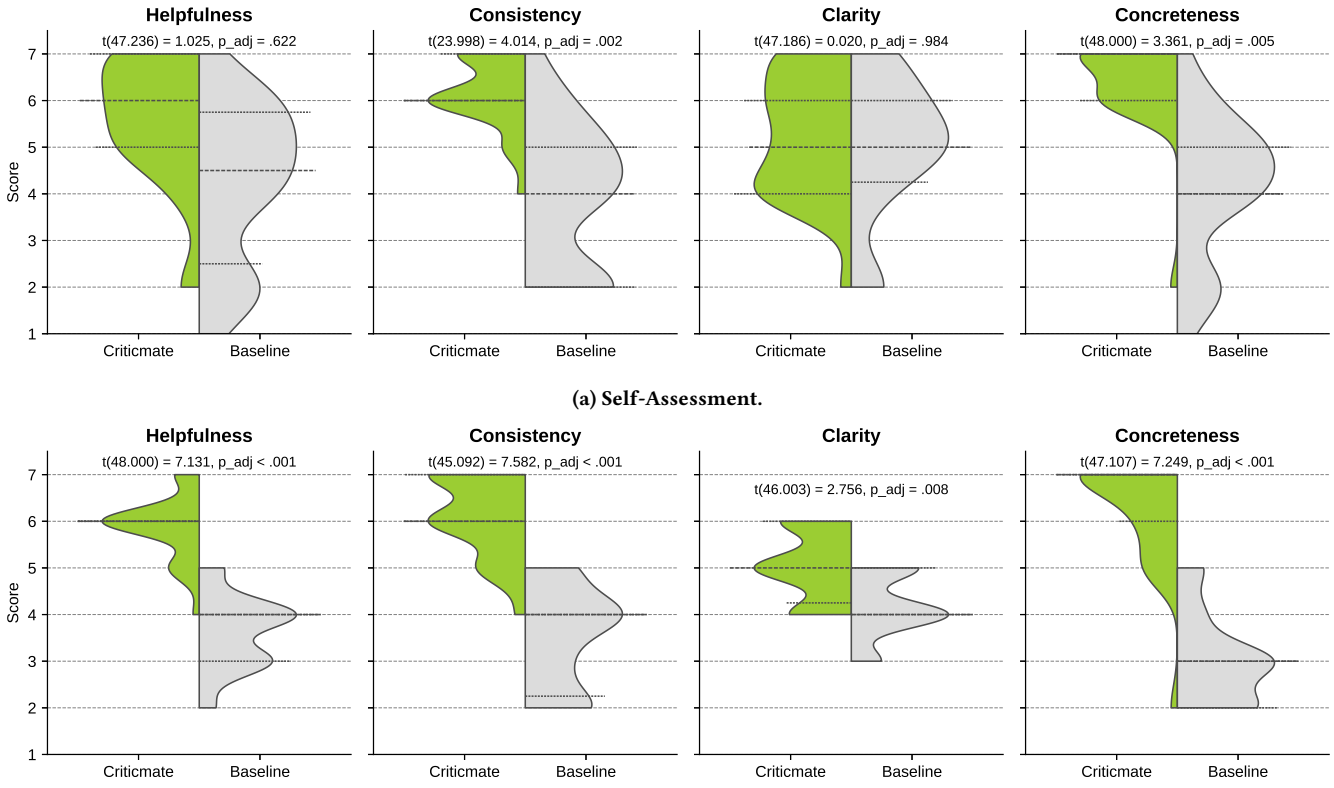
We first report critique-quality ratings, then collaboration experience and stagewise practices. Across all LMMs, we found no reliable main effects of expertise and no expertise × condition interaction; we therefore focus on the fixed effect of collaboration mode (Criticmate vs. Baseline), collapsing across expertise. Full model specifications, including random-effect estimates and alternative codings, are provided in Supplementary Materials, Section S5.

6.6.1 Perceived and Expert-Assessed Critique Quality. Table 3 summarizes the models for critique-quality ratings (self-reported and externally assessed), and Figure 12 visualize the corresponding rating distributions.

Helpfulness. In self-ratings, both systems were perceived as quite helpful for the single-screen evaluation task, with scores near the upper end of the 7-point scale. Criticmate's mean rating exceeded the baseline's, but the difference was not significant after correction, suggesting a ceiling effect: participants felt that either system provided usable feedback. External experts, however, rated Criticmate's critiques as substantially more helpful. They noted that higher-scoring critiques more often referenced heuristics, were grounded in concrete visual evidence (e.g., alignment or spacing), and proposed realistic alternatives, which they saw more frequently with Criticmate (J1, J2). Participants echoed this view, describing Criticmate as offering *"fundamental improvements for this UI"* (I7) and *"concrete guidelines that I could directly apply"* (E7), while still considering the baseline "helpful enough" for a quick pass.

Consistency. For consistency, both self- and external ratings favored Criticmate. Participants perceived Criticmate's critiques as more logically structured (significant positive mode effect), and experts similarly rated them higher. Experts valued feedback that linked observed problems and proposed changes within a clear guideline–problem–solution flow (J2, J3), and penalized fixes disconnected from the UI context (J1). Participants described Criticmate's output as *"reliable because it was based on clear evidence from the screen"* (I12), whereas baseline critiques were seen as *"loosely described without evidence"* (E4).

Clarity. Clarity showed a mixed pattern. Participants' self-ratings did not differ significantly between systems, though Criticmate's mean was slightly higher. Several noted that both systems were "easy enough to read," and that usefulness and specificity mattered more. Baseline summaries were described as *"fast, convenient, and familiar"* (I4, I11). External experts, however, rated Criticmate's critiques significantly clearer, albeit with a smaller effect than for helpfulness or concreteness. They emphasized that Criticmate more



(a) Self-Assessment.

(b) External Assessment.

Figure 12: Split violin plots of critique quality ratings for single-screen UI evaluations, comparing Criticmate and the baseline.

Table 3: Linear mixed-effects model results for critique quality ratings (*: $p_{adj} < .05$, **: $p_{adj} < .01$, *: $p_{adj} < .001$).**

Metric	Coef.	Std. Error	df	t	p_{adj}
Self-assessment					
Helpfulness	0.619	0.604	47.236	1.025	0.622
Consistency**	1.769	0.441	23.998	4.014	0.002
Clarity	0.011	0.575	47.186	0.020	0.984
Concreteness**	1.692	0.503	48.000	3.361	0.005
External assessment					
Helpfulness***	2.154	0.302	48.000	7.131	< .001
Consistency***	2.511	0.331	45.092	7.582	< .001
Clarity**	0.769	0.279	46.003	2.756	0.008
Concreteness***	3.055	0.421	47.107	7.249	< .001

often made *what* was wrong and *why* it mattered explicit in the same passage (J2), while baseline critiques sometimes mixed screen-specific observations with generic advice, making the core message harder to follow.

Concreteness. Concreteness showed the strongest advantage for Criticmate. Participants' self-ratings and external assessments both favored Criticmate, with the largest mode effect among the four dimensions in the expert ratings. Experts rewarded critiques that referenced identifiable UI components, tied issues to the given task, and proposed context-specific remedies (J1–J3); generic advice that could apply to almost any screen received low scores. Intermediates in particular appreciated Criticmate's concrete, screen-anchored feedback, describing it as *"pointing out things I wasn't sure about, but in very screen-specific terms"* (I4). Several participants noted that working through Perception and Comprehension nudged them to treat heuristics as principles instantiated in concrete examples, rather than as abstract checklists.

6.6.2 Collaboration Experience with AI. Figure 13 summarizes collaboration experience ratings; Table 4 reports the corresponding LMM estimates. Again, we observed no reliable expertise effects or interactions.

Engagement. Overall engagement and the Focused Attention subscale were significantly higher for Criticmate. Participants reported being more absorbed and described the staged evaluation as something they could "sink into," working through the screen step by step. The Reward subscale also favored Criticmate: interaction felt more worthwhile and intrinsically rewarding than with the single-pass baseline.

Trust. Trust showed a similarly strong effect. Participants reported substantially higher trust in Criticmate's critiques. Qualitative responses linked this trust to the visibility of intermediate

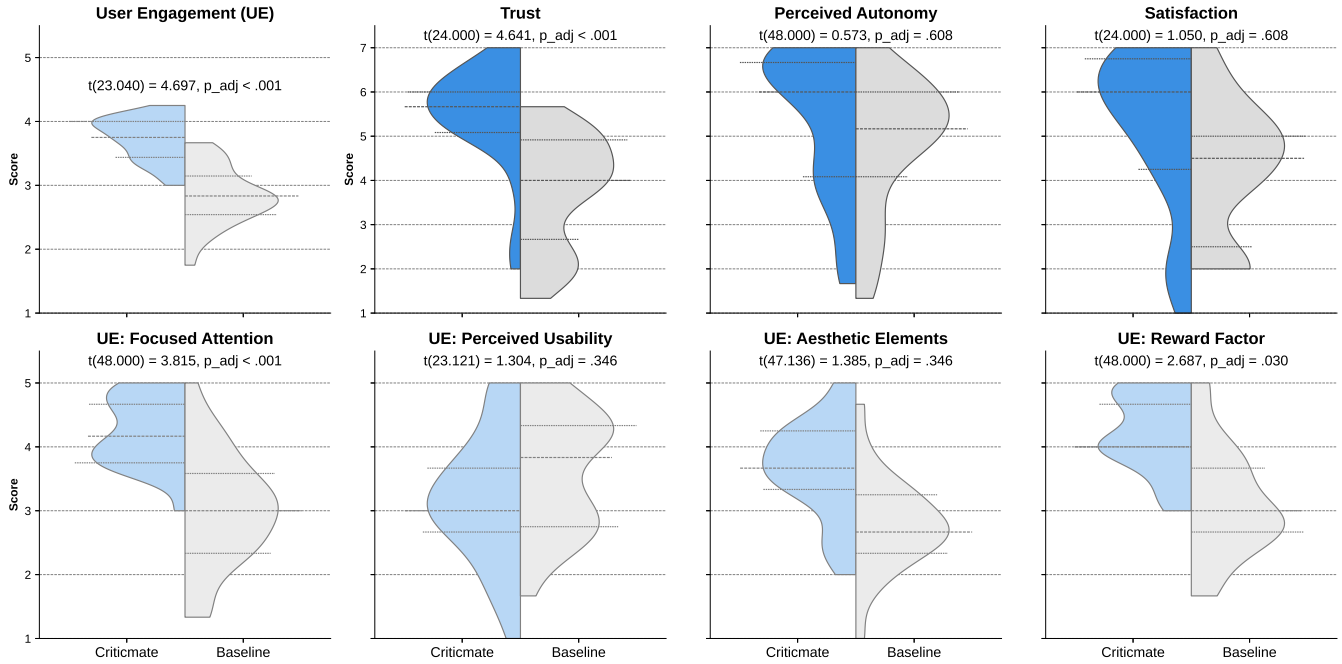


Figure 13: Split violin plots of participant ratings of collaboration experience with Criticmate vs. baseline.

reasoning: *"I could intervene step by step and feel my input was reflected in the outcomes"* (I1, E5); *"it felt like working together to find the best answer for the screen"* (I3, I5); and *"the outcome was convincing because the process was shared"* (I1). Some participants also emphasized Criticmate's potential as a learning tool for heuristic evaluation, especially for less experienced designers: *"for those with less design knowledge, it would be very helpful as a learning resource"* (I7, I8).

Usability, Aesthetic Appeal, Autonomy, and Satisfaction. Perceived usability and aesthetic appeal showed no reliable differences; both interfaces were seen as usable and visually acceptable for this study's tasks. Satisfaction showed a positive but non-significant coefficient, indicating broadly similar overall satisfaction across systems. Open-ended comments help explain this: four participants, including both experts and intermediates, reported being more satisfied with the baseline, citing its speed and simplicity, and several characterized it as *"fast, convenient, and familiar"* (I4, I11). Perceived autonomy likewise did not differ reliably; participants generally felt that both systems allowed them to maintain control and express their preferences in the evaluation process.

Stage-specific reflections further contextualize these ratings. In *Perception*, experts likened Criticmate's detailed parsing to *"reviewing a PRD line by line"* (E10), while intermediates said the structured descriptions made the screen *"easier to understand"* (I4). In *Comprehension*, experts appreciated being able to externalize usually tacit reasoning (E4, E12), and intermediates valued having component functions articulated explicitly (I2). In *Projection*, experts emphasized domain-specific adaptation—*"being able to adjust guidelines for our own domain was impressive"* (E7, E12)—while intermediates benefited from concrete, heuristic-based guidance (I7, I8). At the same time, some participants, especially intermediates, found Criticmate *"relatively unfamiliar and complex"* (I1, I3, I13), and experts noted that the full staged process was not always necessary for simple screens, even if specific stages were useful as complements to existing workflows.

Table 4: Linear mixed-effects model results for collaboration experience (*: $p_{adj} < .05$, **: $p_{adj} < .01$, *: $p_{adj} < .001$).**

Metric	Coef.	Std. Error	df	t	p_{adj}
User Engagement (UE)***	0.703	0.150	23.042	4.697	< .001
UE: Focused Attention***	1.128	0.296	48.000	3.815	< .001
UE: Perceived Usability	0.463	0.355	23.121	1.304	0.346
UE: Aesthetic Elements	0.396	0.286	47.136	1.385	0.346
UE: Reward Factor*	0.795	0.296	48.000	2.687	0.030
Trust***	1.821	0.392	24.000	4.641	< .001
Perceived Autonomy	0.359	0.627	48.000	0.573	0.608
Satisfaction	0.692	0.659	24.000	1.050	0.608

6.6.3 Stagewise Co-Critique Practices. We next examined how participants collaboratively refined AI-generated evaluations across the three stages—Perception, Comprehension, and Projection—within our single-screen UI setting. As shown in Table 5, edits were front-loaded: activity was highest in the early stages and tapered off toward the end. Most interventions targeted component-level details, and participants predominantly edited by adding and modifying content, with deletions playing a smaller but still meaningful role. Within this overall pattern, experts and intermediates expressed distinct, but overlapping, stagewise practices.

Perception: Making the screen fully explicit. In the Perception stage, participants first checked whether Criticmate's structural parsing faithfully reflected what was visible on the single screenshot. Before reasoning about usability, they ensured that all relevant elements were explicitly represented. Participants frequently added missing components (e.g., a back arrow in the header of the podcast screen) and corrected positional or shape descriptions. One intermediate explained, *"I added parts that existed in the section but were not explicitly specified as components"* (I13), while another noted, *"most buttons were simply described as square, so I corrected those parts"* (I7).

Experts went beyond completeness to refine terminology and align it with professional and platform conventions. They often revised terms into more specific and professional terminology, such as revising *"A wide horizontal rectangle"* into *"A button whose width is adjusted to match the left and right padding"* (E3). Log data show that experts made more edits at section level, whereas intermediates concentrated on component rows. Across both groups, Perception was where participants most actively repaired AI misrecognition, treating the section–component map as a useful but imperfect scaffold that needed to be made "fully explicit," as several participants described it, before subsequent stages.

Comprehension: Surfacing and calibrating interpretations. In the Comprehension stage, participants shifted from listing elements to clarifying what those elements meant and did in context. This stage made implicit interpretations explicit: instead of relying on tacit judgments, they articulated why they believed a component had a particular role or state, often revisiting whether that inference was actually warranted from a single static screen.

Intermediates frequently revised AI descriptions that felt too vague or too confident. They enriched generic explanations—for example, updating *"this text block describes the podcast"* to *"this text block describes the* currently playing *podcast episode, highlighted in a green background so users can easily identify what is playing"*—and softened overconfident claims when behavior could not be directly observed. On the hotel-booking screen, one participant changed *"this button navigates to the car search page"* to *"this button likely navigates to a 'Cars' section; based on this screen alone, it may take users to a car search page"*. For many intermediates, Comprehension thus served to calibrate the AI's level of certainty and to anchor functional interpretations tightly to visible cues.

Experts used Comprehension to align visual and functional interpretations with the stated task and broader service logic. They asked whether the inferred functions actually made sense for the scenario and sometimes deleted interpretations that overreached beyond what the single screenshot could support. Compared with Perception, edit rates decreased but remained substantial, with experts more likely to modify or delete entire interpretations and intermediates more often adding clarifications.

Projection: Reframing issues and adapting standards. In the Projection stage, participants consolidated earlier work into evaluative judgments and actionable recommendations. At the guideline step, they transformed generic heuristics into rules that better matched the specific domain and task. Experts frequently rewrote or added criteria to embed domain and business logic—for example, emphasizing login CTAs and error recovery on the hotel screen, or multimodal playback cues and hierarchy on the podcast screen. They described this as *"adapting heuristics to the domain rather than using them as-is"* (E7) and noted that *"domain-specific evaluation criteria are required"* (E11). Intermediates focused more on clarifying and specifying generic guidelines, reinforcing rules around text clarity, icon consistency, and visibility: *"I revised the guidelines to make them easier to understand"* (I8) and *"I applied consistency to text and icons"* (I9). Many reported that seeing guidelines explicitly instantiated for one screen helped them understand how abstract principles translate into concrete checks.

Several experts also used Projection to articulate the structural limitations of working with AI in a single-screen, static setting, with no access to interaction history or flow. They observed that Criticmate's suggestions, like the task framing itself, often reflected only the current UI state without regard to prior or subsequent steps in the flow: *"the AI seemed to consider only the static state and not the existing UI flow; even for humans, that would be tiring, and it did not feel like it understood the service"* (E10). Another expert similarly remarked that the system *"felt like it only considered the static UI state"* but reported that this limitation could be partly mitigated by revising and tightening design guidelines applied in this stage (E13). One expert summarized the constraint more broadly: *"Looking at just one UI and evaluating its problems or generating a critique is inherently difficult"* (E12). While this single-screen focus constrains the scope of our study, it also makes visible the specific breakdowns and collaboration patterns that arise in such settings, providing a concrete foundation for future work that incorporates broader flows and multi-screen context. Projection thus became the place where participants not only adjusted standards, but also negotiated what kinds of judgments are reasonable when only a single static screen is available.

When reviewing identified gaps and proposed fixes, participants treated Criticmate's output as a draft to be reshaped rather than a final verdict. They added missing issues (e.g., touch accessibility or icon interpretability), pruned redundant or low-priority points, and rewrote vague fixes into more actionable, screen-specific recommendations. On the podcast screen, one expert reframed a surface-level comment about menu visibility into a structural critique of hierarchy and placement: *"The color hierarchy for the menu icon is too low, and its position on the far left diverges from what users have learned to expect"* (E3). Another expert (E8) revised a suggestion to *"add a color change effect on hover"* into *"add a color change effect on press"* so the recommendation matched how users interact with mobile UIs in practice. Intermediates often retained the overall issue structure but enriched it with additional reasoning and concrete suggestions, such as unifying colored sections that did not serve a clear purpose (I4).

Log data indicate that experts were active in editing guidelines and pruning or modifying proposed fixes, using Projection to enforce strategic coherence and to compensate for the lack of multi-screen context. Intermediates became most active in the final fix step, contributing a larger share of additions there than experts. For them, Projection functioned as a synthesis space: by the time they reached this stage, the Perception and Comprehension outputs had become a scaffold they could build on, turning AI-generated drafts into critiques that felt more like their own.

Table 5: Stagewise co-critique practices: proportions of participants engaging in edits at each stage and distributions of edit types for experts and intermediates.

Stage		Experts' Edits				Intermediates' Edits			
		Proportion of Participants (%)	Edit Type (%)			Proportion of Participants (%)	Edit Type (%)		
			Added	Deleted	Modified		Added	Deleted	Modified
Perception	**Section**	61.5	53.3	2.7	44.1	30.8	78.9	0.0	21.1
	Component	84.6	45.3	23.0	31.7	69.2	37.5	23.8	38.7
Comprehension	**Component**	84.6	51.9	21.9	26.2	53.8	62.0	3.6	34.4
	Section	38.5	41.2	13.7	45.0	7.7	57.1	28.6	14.3
Projection	**Guideline Setup**	69.2	27.8	0.0	72.2	61.5	63.6	0.0	36.4
	Identified Gap	53.8	33.1	21.9	45.0	38.5	35.5	0.0	64.5
	Proposed Fix	23.1	20.8	37.0	42.2	38.5	66.3	2.7	31.0

7 Discussion

7.1 Stagewise, Process-Oriented Reasoning as a Basis for Collaboration

Effective collaboration depends not only on the quality of AI outputs, but also on how reasoning is organized and exposed. Single-pass workflows—including our baseline—treat critique as a finished product that users can only refine through high-level re-prompts (e.g., "focus more on accessibility"). This interaction pattern mirrors current use of general-purpose LLMs, but it often does not align with the cognitive structure we observed in our formative study: expert evaluators naturally distribute their work across *Perception*, *Comprehension*, and *Projection* [21, 22], weaving together global and local reasoning and gradually moving from surface checks toward root-cause diagnosis.

Criticmate operationalizes a different paradigm: *stagewise co-critique*. The AI follows a Perception–Comprehension–Projection pipeline aligned with this cognitive structure, and each stage yields an explicit, editable artifact. Rather than producing a single opaque pass, evaluation becomes a sequence of partial interpretations that humans can inspect, correct, or extend. This design has two main implications for collaboration.

First, even without human edits, structuring reasoning around stages improves *baseline* behavior. In our ablations, the full stagewise pipeline (CM-Full) outperformed both single-pass baselines and flattened variants on expert similarity, global–local balance, and diversity, despite using the same underlying models and no in-context examples. This suggests that aligning the decomposition of reasoning with human cognitive stages can be as impactful as adding more data: process-oriented prompting can make models more "expert-like" in their standalone critiques.

Second, stage boundaries make *human intervention points* explicit. In Perception, designers inspect and repair the structural parsing of the screen; in Comprehension, they align visual and functional interpretations with task and domain; in Projection, they adapt guidelines, prioritize issues, and reframe root causes. Log data show that edits are front-loaded in Perception and Comprehension and that experts and intermediates engage in different ways (Table 5). Instead of asking designers to reverse-engineer where a single-pass critique went wrong, Criticmate invites them to adjust reasoning while it is still unfolding, using moves that resemble their existing practice.

While Situation Awareness (SA) provided a useful scaffold for structuring Criticmate, it is ultimately only one lens on critique. It foregrounds stepwise perception and reasoning, but gives less attention to social negotiation, historical context, and iterative reframing in design work. Other perspectives—distributed cognition [42], activity theory [50, 66], reflection-in-action [81], and systems thinking [62, 82]—highlight collaboration across actors and artifacts, cultural and organizational constraints, and ecosystem-level effects. *Future work* can build on our stagewise framing by combining SA-style staging with representations of multi-actor critique sessions, organizational goals, and long-term systemic implications.

7.2 From Automation to Co-Critique in UI Evaluation

Most prior work on AI-supported UI evaluation—including our single-pass baseline condition—treats models as *automated judges*: systems that take a screen as input and output a fixed list of issues, with users interacting mainly through coarse-grained, post-hoc prompts to revise it [17, 85, 93]. Our results suggest instead that UI evaluation is better supported as *co-critique*, where humans and AI share responsibility over time by iteratively shaping the model's reasoning, rather than nudging a single, static critique after it has been produced.

Offline, Criticmate's stagewise pipeline produces heuristic critiques that are closer to expert references than zero-shot and few-shot baselines: it achieves higher semantic similarity and coverage, a more balanced mix of global and local issues, and more diverse comments (Figures 9, 10, Tables 1, 2). Yet performance remains imperfect: UI-element overlap is not perfect for all methods, and failures still arise when tasks are ambiguous or UIs are visually underspecified. These limitations echo prior observations that LLM-based critique cannot fully replace human judgment [17, 48].

The user study shows how human intervention on top of this imperfect automation substantially improves overall critique quality. With Criticmate, both participants and external experts rated single-screen heuristic critiques as more consistent and concrete than those from a single-pass baseline, and experts also judged

them more helpful and clearer (Table 3, Figure 12). Participants reported higher engagement and trust (Table 4, Figure 13).

Crucially, Criticmate changed *how* work was divided. The system reliably handled surface-level checks and coverage (e.g., spacing, contrast, basic hierarchy), while designers contributed task context, domain and brand knowledge, and long-term strategy. Intermediates treated this as a scaffold for learning structured heuristic evaluation; experts used it to free attention for higher-level reasoning. At the same time, several experts noted that a full staged workflow can feel heavy for trivial screens, highlighting a depth–speed trade-off and suggesting a need for adaptive staging.

These findings relate to work on automation bias, algorithm aversion, and "appropriate reliance" [5, 32, 44, 87]. Rather than asking whether designers should trust an automated judge, Criticmate makes intermediate reasoning visible and editable, allowing users to decide *when* to rely on or override AI suggestions. In line with research on explainability and trust [41, 51, 74], trust here stems less from anthropomorphic cues than from structured, contestable reasoning. Our contribution is therefore not only a stronger automated evaluator, but a concrete instantiation of *human–AI co-critique* for UI evaluation.

7.3 Design Implications for Human–AI Co-Critique

Our findings suggest several design principles for co-critique systems. Although grounded in single-screen UI evaluation, these implications extend to other domains where interpretability, process alignment, and shared control are important.

Enable intervention during reasoning. Designers intervened most in early stages such as Perception, where errors are frequent and inexpensive to fix. Systems should therefore expose intermediate results as editable artifacts and position evaluation as an interactive workspace rather than a single-pass generator [3, 24].

Structure outputs around familiar reasoning. SA-inspired stages aligned with many designers' mental models and reduced cognitive burden [69], but four participants—including one expert—found the workflow mismatched with their habits. Future systems should offer alternative structures (e.g., reflection-in-action, distributed cognition) or configurable "reasoning templates," so users can choose decompositions that fit their practice.

Connect global and local reasoning. Participants valued Criticmate's ability to relate micro-level flaws (e.g., typography, spacing) to macro-level issues (e.g., hierarchy, layout rhythm). This linkage helped them prioritize fixes and understand cascading effects. Co-critique tools should treat global and local feedback as coupled views over the same reasoning, rather than separate checklists.

Bridge single-screen and multi-screen context. Our current scope is single-screen heuristic evaluation, yet many participants noted that real projects rarely involve isolated screens. Whether a design is appropriate often depends on what came before and what happens next. Future tools should therefore situate single-screen critiques within larger flows—for example, by aggregating issues along a user journey, linking local problems to earlier steps, or letting designers move between screen-level views and higher-level artifacts (e.g., storyboards, service blueprints) while keeping critiques synchronized.

Surface root causes rather than symptoms. In Projection, we grouped surface issues into systemic explanations, which experts valued for supporting long-term improvement. Systems should adopt diagnostic structures that trace repeated symptoms back to missing rules, style guides, or organizational practices, shifting feedback from piecemeal fixes toward strategic guidance.

Make guidelines adaptive and reusable. Editable guideline sets—the "expected standards" in our triplets—were key for aligning evaluation with brand, accessibility, and domain conventions. Treating these standards as reusable design tokens could make evaluation portable across projects and teams and capture organizational knowledge for reuse.

Adapt depth to task and expertise. Experts and intermediates used Criticmate differently: experts concentrated edits in structural recognition, while intermediates intervened most in Projection. Systems should adapt staging depth dynamically—collapsing steps for simple screens, expanding them for complex flows, and offering guided modes for novices alongside streamlined modes for experts—to maintain value across tasks and skill levels.

8 Limitations and Future Work

This work has several limitations that also suggest promising directions for future research.

Scope and generalizability. We deliberately scoped our paradigm and implementation to single-screen, heuristic-style evaluation of mobile UIs. In practice, designers work with multi-screen flows, cross-channel journeys, and long-term use, where concerns such as navigation, temporal patterns, and contextual shifts are central. Our user study further used two stimuli drawn from the higher-density subsets of UICrit and chosen to contrast an information–consumption screen with a task-oriented login screen. Although UI identity was modeled as a random effect in our mixed-effects analyses, generalization to other layouts, domains, device types, and types of critique (e.g., conceptual, socio-technical, or longitudinal critique) remains open. In addition, we focused on critique quality and collaboration experience rather than downstream design outcomes; we did not examine how stagewise co-critique affects iteration processes, final product quality, or alignment with usability, creativity, and organizational goals.

Study design and sample. Our baseline condition approximates how designers currently use general-purpose LLMs for critique: request a one-shot list of issues for a screen, then refine it with natural-language revision prompts. This "single-pass plus revision" pattern is common in practice, but it is not the only plausible comparison. We did not, for example, test baselines that expose only an editable issue list without SA-style staging, or that provide different granularities of intermediate representations. Our participant pool also included only professional designers and intermediates; complete novices, non-design stakeholders, and classroom settings were not studied, even though several participants explicitly highlighted Criticmate's potential as a learning tool. External critique quality was assessed by three professional raters with high inter-rater reliability, yet a larger and more diverse panel (e.g., educators, accessibility specialists, product stakeholders) could surface additional perspectives on what constitutes "good" critique.

System-level and technical limitations. On the system side, errors introduced in Perception and Comprehension can propagate into Projection. Intermediate results allow users to repair these errors and re-run later stages, but improving recognition accuracy and making the impact of edits more transparent would reduce reliance on user corrections and increase efficiency. At present we log YAML snapshots and can diff them for analysis, but the interface does not visualize these diffs or the impact of individual edits for end users. Furthermore, Projection currently uses a non-zero sampling temperature to support ideation, so re-running the same stage after a different sequence of edits may not be strictly deterministic. We treat this limited reproducibility as an acceptable trade-off for richer suggestion diversity in a critique tool, but future work could explore deterministic modes or interface cues that make the consequences of re-running clearer. Finally, Criticmate produces text-only critiques; it does not yet integrate visual annotations, overlays, or direct connections to existing design tools. This limits how easily feedback can be inspected, discussed, and acted upon in professional workflows, and suggests opportunities for tighter integration with design environments, accessibility tooling, and brand systems.

9 Ethics Statement

As described in Sections 3 and 6, participants were recruited on a voluntary basis through professional and academic channels, provided informed consent, and received monetary compensation. No personally identifiable information was collected; all sessions were conducted remotely, recordings were stored securely, and transcripts were anonymized prior to analysis. The procedures involved minimal risk (primarily time and potential fatigue), and session length was capped to mitigate burden.

10 Conclusion

We presented stagewise co-critique, a collaboration paradigm that conceptualizes UI heuristic evaluation as a structured, stagewise process shared between designers and AI. Grounded in a formative study of expert critique practices, we decomposed evaluation into the Situation Awareness stages of Perception, Comprehension, and Projection, and specified what kinds of reasoning and intervention are meaningful at each stage. We operationalized this conceptualization in Criticmate by exposing intermediate reasoning as editable artifacts, enabling designers to correct recognition, inject domain intent, and steer evaluative criteria as critique unfolds.

Our offline evaluation isolates the effect of this stagewise structure on AI behavior itself. Compared to single-pass and flattened variants, the stagewise pipeline produced heuristic critiques that were closer to expert references, more diverse, and better balanced between global and local concerns. At the same time, residual grounding errors and mismatches with expert judgment remain, underscoring that improved structure alone does not eliminate the need for human intervention in UI evaluation.

Our user study complements this analysis by examining what this improved structure affords in human–AI collaboration. When designers engaged with the same stagewise pipeline through Criticmate, they produced critiques that were perceived as more helpful, consistent, and concrete than those generated through a one-pass baseline. Participants reported higher trust and engagement, attributing these gains not to anthropomorphic cues but to the transparency and editability of intermediate reasoning. Root-cause analysis further supported a shift from piecemeal fixes toward more strategic, principle-based guidance, even within the constraints of single-screen inspection.

These findings suggest that process-centric human–AI evaluation tools—those that structure reasoning into explicit stages and invite timely human intervention—can simultaneously strengthen AI performance and collaboration quality. Looking ahead, extending stagewise co-critique beyond single screens to multi-screen flows, cross-device ecosystems, and socially situated critique contexts offers a promising direction for future work. We hope this work motivates a research agenda that moves beyond one-shot judgments toward structured, editable reasoning, advancing evaluation and critique tools that are not only accurate, but also interpretable, adaptable, and grounded in real design practice.

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A Criticmate Output Examples

A.1 Perception

Section Recognition

```
non_app_ui:
  - "Status bar (time, battery, network)"
app_ui:
  Header:
    position: "Top"
    size_shape: "Full width, horizontal bar with back and
     action icons"
```

Component recognition

```
Navigation Tabs:
  Messages Tab:
    position: "Leftmost in the horizontal layout"
    size_shape: "Approximately one-third of the width,
     rectangular"
    sub_components: ["Icon: Envelope", "Text: MESSAGES"]
  ...
```

A.2 Comprehension

Component Analysis

```
Messages Tab:
  visual_characteristics:
    "Leftmost rectangular tab, ~1/3 width, blue envelope
     icon above blue 'MESSAGES', all centered."
  functional_characteristics:
    "Selectable tab for viewing messages; blue visual cue
     indicates it is active."
...
```

Section Analysis

```
Navigation Tabs:
  visual_characteristics:
```

```
    "A full-width horizontal bar with three equal tabs.
     Each tab has a centered icon and label; 'Messages' is
     blue to show it is active."
  functional_characteristics:
    "Allowing users to switch between different views:
     Messages, Alerts, and Connections. The active ..."
...
```

A.3 Projection

Identified Gaps: Section Level

```
global_issues:
  - expected_standard:
      "The expected standard is that the layout should
     maintain a consistent visual hierarchy and alignment
     across sections to facilitate easy navigation and ..."
    identified_gap:
      "In the current design, while each section is
     individually well-aligned, the transition between
     sections lacks a clear visual separation, which ..."

section_issues:
  Navigation Tabs:
    - expected_standard: "The expected standard is that the
     active tab should be clearly highlighted with a ..."
      identified_gap: "In the current design, active and
     inactive tabs mix blue and grey differently, so ..."
  ...
```

Identified Gaps: Component Level

```
Navigation Tabs:
  component_issues:
    "Messages Tab":
      - expected_standard:
          "The expected standard is that the active tab
     should be clearly distinguished using icon, ..."
        identified_gap:
          "In the current design, only the underline and
     icon differ, the label color matches the inactive ..."
    "Alerts Tab":
      - expected_standard:
          "The expected standard is that inactive tabs
     should use a consistently muted style for both ..."
        identified_gap:
          "In the current design, the icon is grey but the
     label is blue, mixing inactive and active signals ... "
...
```

Root Causes Analysis

```
categorized_issues:
  "Color Contrast and Visual Cues":
    root_cause:
      - "Color choices do not provide sufficient ..."
      - "Visual cues for active/inactive states are ..."
      - "Final root cause: Poor color contrast ..."
    issues:
      - component: "Navigation Tabs"
        description: "The active 'MESSAGES' tab is only
     indicated by a thin blue line while all tabs use ..."
      - component: "Profile Section"
        description: "Medium gray text on a pink background
     lacks sufficient contrast for optimal legibility."
  ...
```

Proposed Fixes

```
"Color Contrast and Visual Cues":
  root_cause: "Poor color contrast and ambiguous state
     indicators hinder readability and quick ..."
  individual_fixes:
    - component: "Navigation Tabs"
      issue: "Primary task not visually emphasized."
      final_solution:
        expected_standard:
          "The expected standard is that navigation tabs
     must employ distinct visual indicators to clearly ..."
        identified_gap:
          "In the current design, insufficient
     differentiation in color or style between active ... "
        proposed_fix:
          "Revise the styling for active tabs by using a
     more saturated blue or additional indicators, such ..."
  ...
```

B Study Survey Items

B.1 Perceived Critique Quality

Participants evaluated the quality of the final critique outputs on a 7-point Likert scale (1 = strongly disagree, 7 = strongly agree).

- **Helpfulness**: This critique provides practical help for improving the given UI design.
- **Consistency**: The critique's claims are logically consistent with clear reasoning.
- **Clarity**: The critique is easy to understand and clearly expressed.
- **Concreteness**: The critique is specific to elements of the given UI, rather than general advice.

B.2 Trust

Participants evaluated their trust in the critique process on a 7-point Likert scale (1 = strongly disagree, 7 = strongly agree).

- Using this method to generate a design critique felt safe and trustworthy.
- The information produced through this method was reliable.
- Overall, this method for generating a design critique could be trusted.

B.3 Perceived Autonomy

Participants evaluated their perceived autonomy during the critique process on a 7-point Likert scale (1 = strongly disagree, 7 = strongly agree).

- This method adequately reflected my opinions.
- This method provided critiques according to my standards and preferences.
- I believe I played a leading role in the critique process when using this method.

B.4 Satisfaction

Participants reported their satisfaction with the critique process on a 7-point Likert scale (1 = strongly disagree, 7 = strongly agree).

- I am generally satisfied with this method for generating critiques.

B.5 User Engagement

Participants rated their engagement during the critique process on a 5-point Likert scale (1 = strongly disagree, 5 = strongly agree).

Focused Attention:

- While using this method, I was completely absorbed.
- While using this method, I felt that time passed quickly.
- While using this method, I was deeply engaged.

Perceived Usability:

- While using this method, I felt frustrated.
- While using this method, I felt confused.
- While using this method, I felt mentally fatigued.

Aesthetic Appeal:

- While using this method, I found it attractive.
- While using this method, I found it aesthetically appealing.
- While using this method, it appealed to my visual senses.

Reward Factor:

- While using this method, I felt it was worthwhile.
- While using this method, I felt it was rewarding.
- While using this method, I found it interesting.